**Project Design Phase-I**

| | |
|---|---|
| Date | 28 October 2026 |
| Team ID | Individual Project - Ujjawal Rathod |
| Project Title | Electric Vehicle Market Performance & Insights |
| Maximum Marks | 3 Marks |

## Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

| **Project Overview** | |
|---|---|
| Objective | Convert the combined EV market dataset into a clear, interactive analytics experience that helps stakeholders examine vehicle-category sales, model-year patterns, manufacturers and market share, battery technology, driving range, customer ratings and geography. |
| Scope | The scope covers the two supplied CSV files (about 200,000 training rows and 100,000 test rows), unioned into 300,000 rows and 37 columns; nine Tableau visualizations; interactive filters; a four-point, persona-driven Tableau Story; and a Flask web portal that embeds the published Tableau experience. Forecasting EV resale values or battery degradation is outside the current scope. |
| **Problem Statement** | |
| Description | EV market information is held in large tabular files with many attributes. Manually comparing categories, model years, manufacturers, regions, battery types, driving ranges and customer ratings is time-consuming. Without a unified interactive view and consistent manufacturer market-share calculations, it is difficult to compare segments and support portfolio decisions efficiently. |
| Impact | A consolidated, filterable analytics solution reduces repetitive spreadsheet work, makes category and market patterns easier to explain, supports competitor and regional comparisons, and helps decision-makers explore battery/range and satisfaction trends using a common set of views. |
| **Proposed Solution** | |
| Approach | Combine the two CSV sources in Tableau, validate the analysis fields and use Tableau calculated measures, including FIXED Level of Detail (LOD) logic for manufacturer market share. Build the nine views and interactive filters, organize a four-point Story around the three stakeholder personas, then embed the Tableau Public dashboard/Story in a responsive Python Flask web portal and deploy the portal on Vercel. |

| | |
|---|---|
| Key Features | Nine Tableau views over the unified 300,000-row, 37-column dataset; interactive filters for market region, battery type and model year; FIXED LOD manufacturer market-share logic; a persona-driven Story; and a responsive Flask portal deployed on Vercel with GitHub source control. |

## Resource Requirements

| Resource Type | Description | Specification/Allocation |
|---|---|---|
| **Hardware** | | |
| **Computing Resources** | CPU / compute | Standard 64-bit laptop/desktop; multi-core CPU. No dedicated GPU is required for the Tableau/Flask workflow. |
| **Memory** | RAM | 8 GB RAM minimum; 16 GB recommended for smoother work with Tableau and large CSV files. |
| **Storage** | Disk space | Allow at least 2 GB of free working space for CSV files, Tableau workbook/extracts, application code and documentation; actual need depends on workbook and extract size. |
| **Software** | | |
| **Frameworks** | Analytics and web frameworks | Tableau Public/Desktop for analysis and dashboards; Python 3 with Flask for the web portal; HTML/CSS for responsive presentation; Vercel for deployment. |
| **Libraries** | Libraries / analytics logic | Flask and its required dependencies for the web application; Tableau calculated fields and FIXED LOD calculations for analytical logic. |
| **Development Environment** | IDE and version control | VS Code or another Python IDE; Tableau Public/Desktop; Git and GitHub for source version control; browser for validation and demonstration. |
| **Data** | | |
| **Data** | Source, size and format | ev_market_train.csv (~200,000 rows) and ev_market_test.csv (~100,000 rows), combined to about 300,000 rows across 37 columns. CSV files are used as the data source for the Tableau analysis. |